

EXPERT CONSENSUS DOCUMENT

The Society of Thoracic Surgeons (2025) Expert Consensus Document on Interventions for Screen-Detected Lung Nodules



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ABSTRACT

BACKGROUND Computed tomographic (CT) lung cancer screening (LCS) reduces lung cancer-specific mortality and improves survival. We reviewed contemporary literature to develop consensus recommendations on perioperative quality standards for LCS programs to optimize outcomes.

METHODS The Society of Thoracic Surgeons (STS) Task Force on Lung Cancer Screening and STS Workforce on Evidence Based Surgery convened a multidisciplinary panel of thoracic surgeons, radiation oncologists, and interventional pulmonologists. A comprehensive literature review was conducted using the PICO (Population, Intervention, Comparisons, Outcome) framework. Consensus statements were developed through a modified Delphi process addressing: (1) preoperative biopsy and diagnostic surgical procedures, (2) acceptable rates of complications from diagnostic and therapeutic procedures, and (3) timing of intervention after a suspicious LCS finding, and the role of the multidisciplinary team in patient management. Consensus required ≥75% agreement.

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RESULTS The panel developed 23 consensus statements after 3 Delphi rounds; 20 achieved consensus in the first round. Surgery without tissue diagnosis was acceptable for carefully selected patients, favoring minimally invasive, parenchymal-sparing approaches. Pneumonectomy without diagnosis was unanimously rejected. Programs should track benign resection rates. Acceptable complication benchmarks included pneumothorax <5%, hemoptysis <2%, and mortality <1% for bronchoscopic biopsy; and surgical morbidity <10% and 30-day mortality <1%, per STS database standards. Definitive resection should occur within 12 weeks of the inciting imaging study. Multidisciplinary teams should include thoracic surgery, oncology, pulmonology, and radiology. Preoperative pulmonary rehabilitation and smoking cessation were emphasized.

CONCLUSIONS This STS consensus defines perioperative quality standards for CT LCS programs, supporting shared decision-making, multidisciplinary care, and quality improvement.

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EXECUTIVE SUMMARY

Computed tomographic (CT) lung cancer screening (LCS) has proven efficacy in reducing lung cancer-specific mortality, yet variability in downstream diagnostic and surgical management poses risks to patient safety, quality, and confidence in LCS. In recognition of this, The Society of Thoracic Surgeons (STS) convened a multidisciplinary panel of thoracic surgeons, radiation oncologists, and interventional pulmonologists to establish evidence-informed, consensus-based standards for perioperative care and quality assurance in the management of screen-detected lung nodules. Through a structured literature review and a modified Delphi process, the panel developed 23 consensus statements addressing preoperative biopsy and diagnostic surgical procedures, acceptable complication thresholds, and timeliness of intervention, emphasizing the importance of multidisciplinary coordination throughout care delivery. This Expert Consensus Document (ECD) provides a framework for surgical and perioperative quality standards that CT LCS programs should strive to achieve. Collectively, these consensus statements underscore that maintaining rigorous quality metrics, data tracking, and multidisciplinary collaboration are essential to ensure that the life-saving potential of LCS is realized through safe, efficient, and patient-centered care.

INTRODUCTION

Lung cancer is the leading cause of cancer mortality worldwide, and low-dose CT screening

demonstrably reduces lung cancer-specific mortality through earlier detection of disease.^{1,2} With expanded access to lung cancer screening (LCS), the detection of both benign and malignant lung nodules has increased substantially.³ Many of these nodules require diagnostic evaluation or therapeutic intervention.

Despite clear survival benefits, however, concerns persist regarding potential downstream harms of screening, including unnecessary invasive procedures, radiation exposure, and overtreatment.⁴ These concerns are often amplified by inconsistent definitions and methodologic flaws in the literature that may overestimate complication rates and misrepresent the risks associated with biopsy or surgery, which have steadily fallen over time.⁵ This misinformation may undermine confidence in screening programs and deters patients from participation, despite ample data demonstrating LCS is safe, effective, and life-saving.

The establishment of clear, consensus-based benchmarks is essential to ensure that the mortality benefit of screening is not offset by avoidable procedural risk or variability in care. This need will likely grow in importance as eligibility criteria for cancer screening broadens—including consideration of total years of smoking rather than pack-year thresholds, removal of the 15-year quit-time limitation, and inclusion of select higher-risk never-smokers, such as Asian women.^{6,7}

The likely increase in screening will drive an increase in the number of downstream procedures and enhance the importance of standardized benchmarks for biopsy and surgical management.

Abbreviations and Acronyms

CALGB/Alliance = Cancer and Leukemia Group B/Alliance
CT = computed tomography
ECD = Expert Consensus Document
LCS = lung cancer screening
Lung-RADS = Lung Imaging Reporting and Data System
MDM = multidisciplinary meeting (tumor board)
NSCLC = non-small cell lung cancer
PET = positron emission tomography
RATS = robotic-assisted thoracic surgery
SBRT = stereotactic body radiotherapy
STS = The Society of Thoracic Surgeons
STS-GTSD = The Society of Thoracic Surgeons General Thoracic Surgery Database
TTNB = transthoracic needle biopsy
TTS = time to surgery
VATS = video-assisted thoracoscopic surgery

Clear consensus guidance is therefore critical to safeguard quality, maintain public confidence, and ensure that expanded screening translates into survival improvements.

In recognition of this need, the STS Workforce on Evidence-Based Surgery and the STS Task Force on Lung Cancer Screening convened a multidisciplinary panel to develop an ECD. This document focuses specifically on biopsy and surgical interventions for lung nodules discovered through CT screening. Using a structured review of contemporary literature and a modified Delphi process, the group formulated consensus statements addressing (1) preoperative biopsy and diagnostic surgical procedures, (2) acceptable complication thresholds for diagnostic and therapeutic interventions, and (3) the timing and multidisciplinary coordination of care for screen-detected nodules.

The goal of this consensus document is to provide thoracic surgeons, multidisciplinary screening programs, health systems, patients, and other stakeholders with practical, evidence-informed benchmarks that promote high-quality, patient-centered care and reduce unwarranted variability in the management of screen-detected lung nodules.

METHODOLOGY

In 2024, the STS Task Forces on Lung Cancer Screening and Evidence-Based General Thoracic Surgery convened a team of multidisciplinary experts to develop an ECD on detection and surgical management of screen-detected lung nodules. Participants were selected for their clinical expertise in early detection and management of lung cancer. All members submitted conflict of

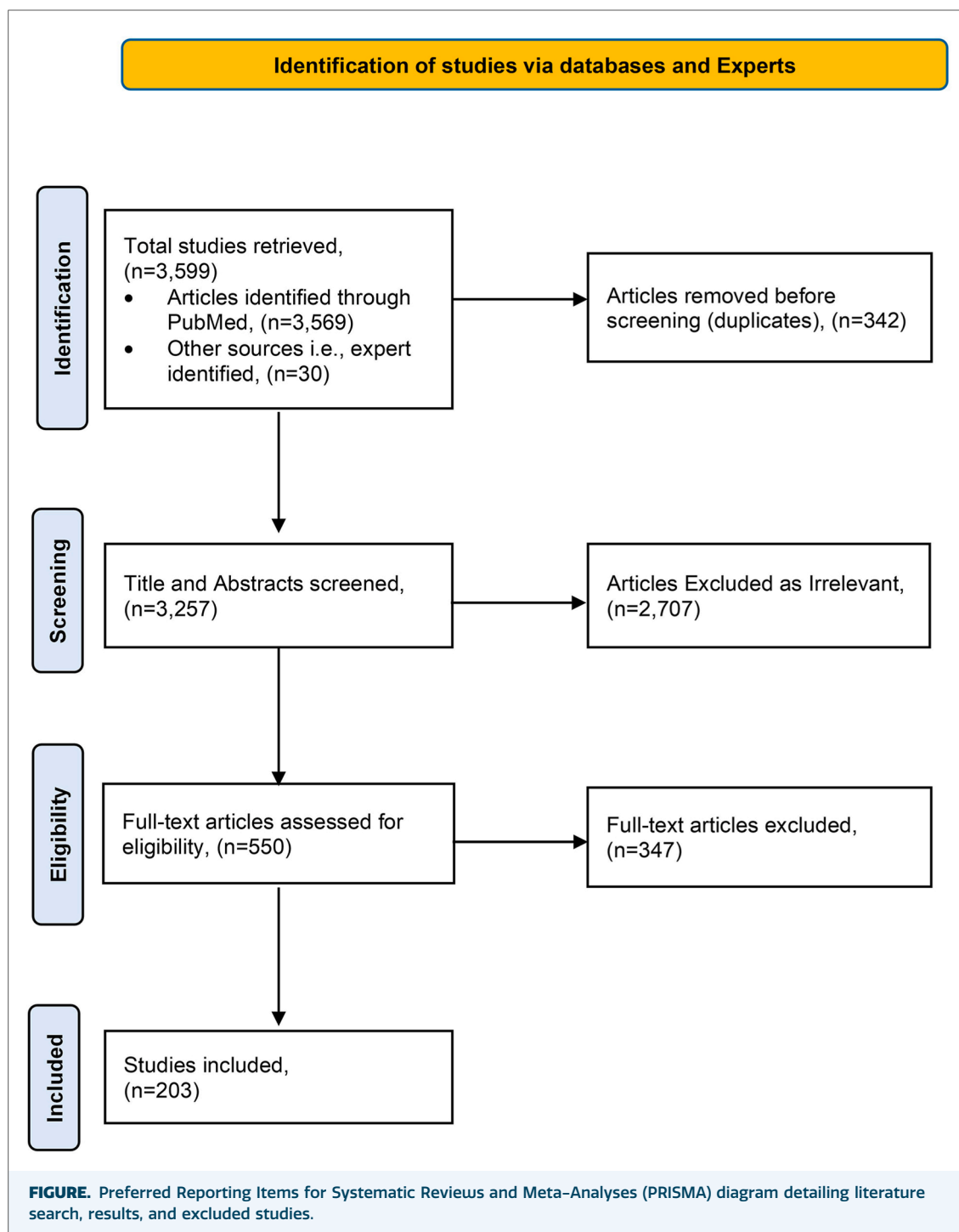
interest disclosures, which were reviewed by the project chair and STS leadership. None of the disclosures constituted a material conflict within the scope or intent of the ECD. The disclosures upon review did not warrant any modification in participation. Clinical questions were developed using the PICO (Population, Intervention, Comparison, Outcomes) framework.⁸ Research questions addressed 3 themes: (1) preoperative biopsy and diagnostic surgical procedures, (2) acceptable rates of complications from diagnostic and therapeutic procedures, and (3) timing of intervention after a suspicious LCS finding and the role of the multidisciplinary team in patient management.

In collaboration with the writing group, a medical librarian designed comprehensive search strategies in PubMed covering human studies in English, January 2011 through August 2024. Additional studies were identified through expert review and manual screening of reference lists. Study design was unrestricted, excluding only gray literature, abstracts, dissertations, editorials, and animal studies. Inclusion and exclusion criteria are detailed in [Supplemental Table 1](#). The full PubMed search strategy is provided in [Supplemental Table 2](#).

After duplicates were removed, all records were screened in 2 phases: titles and abstracts, followed by full-text review against predefined inclusion and exclusion criteria. At the full-text stage, 2 independent reviewers evaluated each study, with any disagreements resolved through discussion.

The initial database and hand searches yielded 3316 unique records after duplicates were removed. After the 2-phase screening, 203 studies were included to inform consensus statement development. A Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram⁹ summarizing the process is shown in the [Figure](#).

Consensus statements were developed through a modified Delphi process consistent with Institute of Medicine guidelines.¹⁰ Statements were rated on a 5-point Likert scale (1 = strongly disagree, 5 = strongly agree) and finalized when ≥75% agreement (selected “agree” or “strongly agree”) was achieved.¹¹ The panel completed 3 rounds of anonymous voting, with full member participation. Detailed voting results are provided in [Supplemental Table 3](#). The final document underwent review and approval by the STS Workforce on Evidence Based Surgery, the Council on Quality & Research, and the STS Executive Committee, and was circulated for a 2-week public comment period before publication.



RESULTS

PREOPERATIVE BIOPSY AND DIAGNOSTIC SURGICAL PROCEDURES. **Statement 1:** In carefully selected patients with a LCS-detected lung nodule, it is acceptable to proceed directly to the operating room for lung resection without a preoperative diagnosis.

Management of screen-detected findings can be complex and requires comprehensive evaluation of radiographic imaging characteristics and patient-specific clinical risk factors.¹² Biopsy is often indicated for lesions at risk of malignancy; however, selected patients may appropriately undergo upfront surgical resection without a preoperative diagnosis, especially in those with

a high probability of malignancy and a nodule amenable to wedge resection. Intraoperative diagnosis can avoid added risk, cost, a false-negative result on the biopsy specimen, and delay in treatment from an invasive percutaneous or bronchoscopic biopsy in select cases.

Whenever possible, lobectomy should not be performed without an established diagnosis of lung cancer. Therefore, before proceeding to the operating room, it is appropriate to ensure that (1) the tumor is clinical stage I based on appropriate imaging, (2) the diagnosis can be obtained in the operating room before anatomic resection, (3) the preoperative probability of malignancy is sufficiently high to justify operative intervention without a preoperative diagnosis, and (4) the patient can tolerate curative resection.

Generally, the tumor must be in the peripheral third of the lung. If centrally located and a diagnosis is unlikely to be possible without lobectomy, then multidisciplinary board discussion and consideration for preoperative attempts at biopsy are strongly recommended before an operation. Considering lobectomy without diagnosis is reasonable if prior attempts at biopsy have been unsuccessful or when growth trajectory is strongly consistent with lung cancer.

The American College of Chest Physicians and the British Thoracic Society recommend direct referral to surgery without a preoperative biopsy in patients with lung nodules >8 mm with an estimated probability of malignancy >65% based on validated nodule malignancy risk prediction models.^{13,14} Lung nodule cancer prediction models (eg, Brock University Cancer Prediction Equation, Mayo Clinic Model) are available but have not been shown to be more accurate than experienced judgement. Multidisciplinary input may be useful in determining which patients have a sufficiently high suspicion of malignancy that warrants surgical resection in lieu of preoperative biopsy.

In areas where endemic fungal infections, such as histoplasmosis and coccidioidomycosis, may mimic lung cancer on CT and positron emission tomography (PET)/CT, preoperative biopsy should be considered before an operation, unless growth on serial CT scans is highly consistent with malignancy.¹⁵

Statement 2: In patients proceeding to the operating room without a pathologic diagnosis in LCS programs, diagnostic resection should be performed using minimally invasive techniques.

Minimizing morbidity and mortality is paramount, particularly for diagnostic resections. For

patients proceeding to the operating room without a preoperative diagnosis, minimally invasive techniques are preferred. Video-assisted thoracoscopic surgery (VATS) or robotic-assisted thoracoscopic surgery (RATS) are both acceptable and have several advantages over thoracotomy.^{16,17} In addition to retrospective data from several large national database studies that have consistently demonstrated the short-term benefits of thoracoscopic techniques, the Video-Assisted Thoracoscopic or Open Lobectomy (VIOLET) trial established minimally invasive lung resection as the standard of care in the management of early-stage lung cancer. With a significantly better physical function score at 5 weeks, reduced hospital length of stay, and comparable 1-year survival, VATS is supported by high-level evidence.¹⁸

As such, the use of minimally invasive resection has markedly increased over time¹⁹ and can be applied to more advanced cancers in appropriately selected patients.²⁰ Thoracotomy should be avoided when resection is initially performed for the purpose of obtaining a diagnosis, and the use of adjunct techniques, such as preoperative fiducial placement, intraoperative marking, and 3-dimensional imaging, can be considered to aid in successfully completing a minimally invasive operation.

Statement 3: When proceeding to the operating room without a pathologic diagnosis for a finding discovered through LCS, diagnostic resection should aim to maximally preserve lung parenchyma through a wedge or sublobar resection, avoiding larger anatomic resections.

Consensus was achieved in support of diagnostic sublobar lung resection over lobectomy. Although patients selected for upfront resection typically have high preoperative risk of malignancy, nevertheless, some will have benign pathology, and priority should be given to minimizing the extent of parenchymal resection. Ideally, wedge resection should be performed, given the lower perioperative risks associated with this procedure.²¹ Segmentectomy may be considered for lesions that are deeper within the lung parenchyma; however, more extensive anatomic resections, including diagnostic lobectomy, should be avoided.

Statement 4: Pneumonectomy should not be performed in patients screened in an LCS program without a preoperative diagnosis.

Expert consensus strongly supports mandatory tissue confirmation of malignancy before pneumonectomy. Pneumonectomy is associated with

significantly higher perioperative morbidity, mortality, and long-term functional impairment compared with lobectomy or sublobar resection.^{22,23} Performing pneumonectomy without a tissue diagnosis introduces unjustifiable risk, even when the preoperative probability of malignancy is high. Moreover, when pneumonectomy is under consideration for a patient with suspected lung cancer, multidisciplinary review should be obtained to determine whether neoadjuvant therapies should be offered before surgery and to review nonsurgical options.

Statement 5: In patients with moderate or high operative risk, preoperative biopsy is preferable to diagnostic wedge resection.

Statement 6: In patients with moderate or high operative risk and nondiagnostic preoperative biopsy, it is reasonable to attempt an intraoperative needle biopsy (including intraoperative bronchoscopic biopsy) before proceeding to diagnostic resection.

Resection for benign disease is the risk of proceeding directly to surgery without preoperative tissue diagnosis. Currently, the benign resection rate among individuals participating in LCS programs ranges from 3% to 15% in contemporary reports.^{24–27} For patients with lung nodules and who are at moderate or high operative risk, any diagnostic strategy must prioritize biopsy before resection, given that complications from surgery for these higher-risk populations are more common, severe, and rarely fatal.^{28,29} Strategies to reduce the benign resection rate should be used, including navigational bronchoscopy or CT-guided fine needle aspiration. Both approaches have a diagnostic accuracy >70% with low complication rates.³⁰ If a specific benign diagnosis is established, then operations can be avoided.

Preoperative biopsy specimens can be nondiagnostic in ~15% to 30% of cases.³¹ If a preoperative biopsy specimen is nondiagnostic, attempting an intraoperative needle biopsy (including intraoperative bronchoscopic biopsy) is reasonable before proceeding to diagnostic resection. Samples obtained from intraoperative biopsies are associated with high sensitivity (>80%) and specificity (>98%).³² However, before any operation, the surgeon should have a plan formulated in the event of a nondiagnostic intraoperative biopsy specimen. In the case of a moderate- or high-risk operative patient, this would involve wedge resection with or without nodal sampling alone or segmentectomy. If operative risk is considered significant and the

probability of malignancy is high, empiric stereotactic body radiotherapy (SBRT) to the lesion in question should be discussed in a multidisciplinary tumor board setting as an alternative approach.

Statement 7: The incidence of lung resection for benign disease in patients screened in a LCS program should be less than 10%.

Statement 8: LCS programs should track their rate of resection for benign disease as an ongoing quality metric.

Despite clear reduction in lung cancer-specific mortality, concerns regarding overdiagnosis and unnecessary interventions represent significant barriers to LCS compliance. The rate of screened patients undergoing surgery for benign disease has been a particular focus.

In the National Lung Screening Trial (NLST) and Netherlands Leuven Screening Onderzoek (NELSON) trials, which enrolled patients between 2002 and 2004 and between 2004 and 2008, respectively, benign resection rates ranged from 24% to 28%.^{1,33} Subsequent studies have shown substantially reduced rates of surgery for benign disease in LCS. A retrospective analysis of 163 patients undergoing surgery after LCS at United States International Early Lung Cancer Action Program (I-ELCAP) sites between 2006 and 2011 yielded a benign resection rate of 10%.²⁵ Smaller single-center studies, including patients undergoing surgery between 2012 and 2021 at Lahey Hospital, Mass General Brigham, and Virginia Mason Medical Center, reported benign resection rates of 9% to 17%.^{24,27,34,35} More recently, the benign resection rate in the Manchester Lung Health Check Pilot, which included patients undergoing surgery in 2016 and 2017, was 2.5%.³⁶

This expert consensus panel agreed that a benign resection rate of <10%, for all diagnostic and therapeutic resections, is the appropriate aspirational benchmark for LCS programs. The introduction of robotic-assisted bronchoscopy, which allows for high diagnostic yield even for small peripheral nodules, along with improved imaging technology and operative techniques, help minimize the rate of benign resection.²⁵ It is incumbent on screening programs to track the rate of surgery for benign diagnoses to reduce the exposure of screened patients to avoidable morbidity and health care costs. Ongoing review and quality assurance may identify the need to pursue improved multidisciplinary review and increased preoperative diagnostic methods if the rate of benign resections is >10% within a screening program.

ACCEPTABLE RATES OF COMPLICATIONS FROM DIAGNOSTIC AND THERAPEUTIC PROCEDURES. Statement 9: In patients who require preoperative lymph node staging, the selected diagnostic approach should minimize the total number of procedures necessary to diagnose the lesion and stage the mediastinum.

When a screening CT scan shows an abnormality, all effort should be made to obtain a diagnosis and stage in a timely fashion with minimum risk of morbidity. Scans showing abnormal results should be reviewed by a multidisciplinary team and a plan made to obtain tissue and complete staging in the fewest procedures. With the advent of advanced bronchoscopic techniques, such as robotic navigational bronchoscopy, a patient can undergo biopsy of a pulmonary nodule and concurrent nodal staging using endobronchial ultrasound under the same anesthetic event. However, this may require coordination of resources, logistics, equipment, and expertise.^{37–39} For small peripheral lesions with no abnormal adenopathy noted on CT, PET imaging along with a specimen from a bronchoscopic biopsy or transthoracic needle biopsy may be sufficient to diagnose and clinically stage the patient before definitive treatment.

Statement 10: LCS programs should track the complication rate after diagnostic procedures and perform ongoing quality improvement to minimize this rate.

Complication rates vary widely among reports of interventions after LCS. Carefully tracking a program's complication rate is crucial to inform quality improvement initiatives. In 2025, the International Association for the Study of Lung Cancer published a Delphi consensus document outlining recommended quality metrics for LCS programs.⁴⁰ Complication tracking from diagnostic or therapeutic procedures is considered an essential quality metric. If higher than desired rates of complications are noted within a program, more frequent assessment and targeted interventions should be performed to help rectify this.

Statement 11: The rate of pneumothorax requiring intervention after diagnostic procedures for LCS-detected lung nodules should be <5%.

The 2 main nonsurgical biopsy techniques are transthoracic needle biopsy (TTNB) and bronchoscopic biopsy. Although level 1 data are sparse, both are associated with a very low likelihood of pneumothorax requiring invasive intervention. With TTNB, postprocedure pneumothorax requiring intervention (including tube

thoracostomy and surgery) occurs in ~5%.⁴¹ TTNB postprocedure morbidity decreases when a biopsy is performed on smaller, more peripheral lesions and when a shorter needle length is used. However, postprocedure morbidity is not related to the diameter of the needle.⁴²

In large retrospective as well as prospective, multicenter, cohort studies of electromagnetic navigation-guided bronchoscopic biopsy for the diagnosis of peripheral lung lesions, the overall incidence of pneumothorax ranges from 1.1% to 4.7%, with an even lower incidence of pneumothorax requiring intervention.^{43,44} Studies of diagnostic bronchoscopy using more sophisticated methods demonstrate similar results, with the incidence of pneumothorax requiring intervention after robotic-assisted bronchoscopy ranging from 0% to 6.25%⁴⁵ and the incidence after advanced imaging-guided bronchoscopy ranging from 0.5% to 2.1%.^{46–50} A recent randomized trial reported a pneumothorax rate needing intervention of 0.8% in the robotic navigational cohort and 11.5% in the TTNB cohort.³⁰

Regardless of approach, LCS programs should track the postbiopsy pneumothorax rate, and the multidisciplinary team should strive to minimize this complication. This panel concluded that a <5% pneumothorax rate after invasive diagnostic nonoperative biopsy is an appropriate quality benchmark.

Statement 12: The rate of clinically significant hemoptysis requiring any intervention after bronchoscopic biopsy of LCS-detected lung nodules should not exceed 2%.

Despite being one of the more common adverse outcomes of peripheral bronchoscopy, clinically significant hemoptysis is rare, and rates of hemoptysis requiring an endoscopic or endovascular intervention or unplanned hospitalization are even less common. In the prospective, multicenter NAVIGATE (Clinical Evaluation of superDimension™ Navigation System for Electromagnetic Navigation Bronchoscopy™) study of electromagnetic navigational bronchoscopy, the incidence of hemoptysis requiring any intervention was 1.7%, and an even lower incidence of <0.01% was observed in a recent meta-analysis of diagnostic outcomes and complications of robotic-assisted bronchoscopy.^{43,45} Although comparative studies are lacking, the addition of imaging guidance to peripheral bronchoscopic procedures is associated with a similarly low incidence of hemoptysis, with both prospective

and retrospective studies reporting rates of 0% to 1.1%.⁴⁶⁻⁴⁹

Statement 13: Mortality and major morbidity after bronchoscopic biopsy should be <1%.

Cardiovascular, pulmonary, and anesthetic-related complications, including 30-day mortality, are extremely uncommon after bronchoscopic biopsy; however, practitioners should be aware of these very rare but severe complications in the absence of rigorous data. Periprocedural mortality is exceedingly uncommon, with no deaths occurring in most studies. Of 1388 patients who underwent peripheral bronchoscopic biopsy in the NAVIGATE study, 1 anesthetic-related death and no procedure-related deaths occurred.⁴³ The incidence of cardiovascular or pulmonary complications—including acute exacerbations of chronic obstructive pulmonary disease or acute respiratory failure, myocardial infarction, cardiac arrest, and gas embolism—is also very low. In a retrospective cohort study conducted at Veterans Affairs hospitals, acute respiratory failure occurred in 3.1% of cases, with all other unanticipated pulmonary or cardiovascular events occurring at rates not exceeding 1.1%.⁴⁴ In studies of robotic-assisted and imaging-guided bronchoscopy, pooled incidences of all potential cardiopulmonary complications are generally <2%.⁴⁵⁻⁴⁹

Statement 14: Sublobar resection with lymph node assessment for a LCS-detected non-small cell lung cancer (NSCLC) should be associated with a 30-day mortality <1%, consistent with STS General Thoracic Surgery Database (GTSD) rates.

In patients with LCS-detected NSCLC, sublobar resection with lymph node assessment is safe and associated with excellent perioperative outcomes. Reports from the STS GTSD indicate that wedge resection and segmentectomy combined with mediastinal lymph node evaluation are both associated with a <1% 30-day mortality rate in early-stage disease.^{51,52} These results have also been observed in large, randomized trials such as the Cancer and Leukemia Group B (CALGB)/Alliance 140503 Trial in which sublobar resection for peripheral node-negative NSCLC <2 cm was associated with a 0.6% 30-day mortality.⁵³ These findings have been further corroborated by reports from national data sets such as American College of Surgeons National Surgical Quality Improvement Program, as well as in those from both single and multi-institutional series, in which 30-day mortality rates ranged between 0.6% and 0.7% after sublobar resection in well-selected patients.⁵⁴ These data reaffirm the perioperative safety of

sublobar resection with nodal staging in the management of LCS-detected NSCLC.

Statement 15: Overall 30-day major morbidity after sublobar resection with lymph node assessment for NSCLC should be <6%, consistent with STS GTSD rates.

The STS GTSD defines major morbidity as ≥ 1 postoperative events comprising the composite risk models for lobectomy, including pneumonia, acute respiratory distress syndrome, bronchopleural fistula, pulmonary embolus, initial ventilator support >48 hours, reintubation/respiratory failure, tracheostomy, myocardial infarction, and unexpected return to the operating room. STS GTSD reports major morbidity rates are 4.6% to 5.5% for segmentectomy and 5.2% for wedge.^{55,56} These real-world registry data are consistent with outcomes from randomized trials, such as CALGB/Alliance 140503, which reported grade 3 or higher adverse event rates of 14% for sublobar resection; this includes all major complications, in addition to those meeting strict STS definitions.⁵⁷

In the 2023 ECD on staging and multidisciplinary management of patients with early-stage NSCLC, the American Association for Thoracic Surgery authors reference the STS GTSD as the benchmark for perioperative outcomes in early-stage NSCLC, supporting the <6% major morbidity threshold for sublobar resection and quality assurance.⁵⁸

Statement 16: Lobectomy with lymph node assessment for a LCS-detected NSCLC should have a 30-day mortality of <1%, consistent with STS GTSD rates.

Statement 17: Overall 30-day major morbidity after lobectomy with lymph node assessment for a LCS-detected NSCLC should be <10%, consistent with STS GTSD rates.

When undertaking surgical resection of a screen-detected NSCLC, lobectomy with lymph node assessment may be an appropriate option both for centrally positioned tumors of any size and for many tumors >2 cm in diameter regardless of their location in the lobe. In the most recent data formally published by the STS GTSD, the 30-day in-hospital mortality rate was 0.7%, operative mortality (any in-hospital mortality beyond 30 days) after lobectomy was 1.1%, and the major morbidity rate was 9.5%.²¹ NSCLCs identified on CT screening have an ~60% chance of being in stage I at the time of diagnosis—with a substantial reduction in stage III cases vs chest radiography-screened controls.¹ Given more operations are being performed on earlier-stage disease, we believe that the morbidity and mortality rates after lobectomy for

screen-detected NSCLC will be even lower than those quoted in the STS GTSD.

Statement 18: Minimally invasive surgery should be preferentially performed over thoracotomy for lung resections for NSCLC discovered during LCS.

The vast majority of lobectomies performed for screen-detected NSCLCs are achievable by minimally invasive techniques (VATS or RATS). By 2020 in the STS GTSD, 80% of all lobectomies for lung cancer (including all resectable stages) were being approached by minimally invasive techniques.²¹ Level I, randomized data demonstrate that lobectomy by VATS vs thoracotomy confers better recovery of physical function early after surgery,¹⁸ and in the STS GTSD lung cancer resection risk model, thoracotomy conferred an 87% higher risk of mortality vs VATS and RATS approaches and a 51% higher risk of major morbidity.⁵⁹ Other studies from large databases using propensity matching or multivariate models demonstrate fewer complications,⁶⁰ shorter lengths of stay,^{61,62} and similar long-term survival^{61,62} when minimally invasive techniques are used. Even in a selected group of patients with severely reduced pulmonary function (postoperative predicted forced expiratory volume in 1 second <40%), postoperative mortality was only 0.7% in an STS GTSD analysis.⁶³ Thus, unless technical challenges render minimally invasive techniques impossible or unsafe in particular cases, minimally invasive techniques can and should be preferentially performed.

TIMING OF INTERVENTION FOR SCREEN-DETECTED NODULES AND ROLE OF THE MULTIDISCIPLINARY TEAM. Statement 19: In patients with LCS-detected nodules, definitive surgical resection with intent to cure should occur within 12 weeks from the CT scan that prompted intervention.

The expert panel agreed that definitive curative intent surgical resection should occur within 12 weeks from the diagnosis date, defined as date of the CT scan that prompted intervention. Previous literature, including a recent systematic review, on timely surgical intervention for NSCLC has shown that prompt time to surgical intervention is advantageous for early-stage lung cancer.⁶⁴ However, these studies reported varying time to surgery (TTS) intervals, ranging from 21 days to >8 weeks from the diagnostic date. Further complicating matters, these studies used varying diagnostic definitions, including time from CT or PET/CT scan or biopsy specimen-proven histologic diagnosis.^{65,66}

In 2021, Heiden and colleagues⁶⁷ helped clarify an optimal diagnostic date (CT scan with suggestive findings that prompted intervention) from which to measure time point 0 to accurately assess TTS studies and evaluate the optimal TTS for improved outcomes. The authors showed that in a United States Department of Veterans Affairs population of stage I NSCLC patients, TTS >12 weeks from the CT scan that prompted intervention was associated with increased recurrence and worsened overall survival. In addition, a subsequent follow-up study showed that TTS >12 weeks, was correlated with worse recurrence-free and overall survival.^{65,68}

More recently (2025), Tupper and colleagues⁶⁹ reported TTS >8 weeks from the CT scan that prompted intervention for early-stage NSCLC was associated with increased 1-year recurrence and 5-year mortality. In addition, they reported that for every 2-week delay in surgical intervention beyond 8 weeks from detection, 1-year recurrence and 5-year mortality both increased incrementally. Our expert panel acknowledges a lack of high-quality data to define the optimal time to intervention for screen-detected nodules. As such, the overall group consensus was to allow for less stringent recommendations of TTS <12 weeks as the minimum requirement to perform appropriate timely surgery for LCS-detected nodules.

Statement 20: In high operative-risk candidates, structured, pretreatment pulmonary rehabilitation before defining surgical candidacy may be appropriate, within the 12-week treatment window.

A 2022 Cochrane review by Granger and Cavalheri⁷⁰ and Cavalheri and Granger⁷¹ including 10 randomized trials, examined the role of preoperative exercise training for individuals who were to undergo a planned surgical resection for lung cancer. The study concluded with a high degree of certainty that preoperative exercise training resulted in a large reduction in risk of postoperative pulmonary complications (risk ratio, 0.45; 95% CI, 0.33–0.61). This translated into a 20% reduction in pulmonary complications for participants. Interestingly, there was no difference in outcomes between participants in programs that were <2 weeks in duration compared with those that were >2 weeks in duration. Furthermore, there was moderate certainty regarding reduction in length of stay (–2.24 days; 95% CI, –3.64 to 0.85 days) with preoperative exercise programs.

No difference was found with postoperative mortality between groups. A 2025 prospective, multicenter randomized controlled trial has more recently demonstrated nearly identical results.⁷²

Despite the strong evidence favoring preoperative exercise programs, they have not been widely embraced. This is attributed to concerns regarding delays in treatment, logistical obstacles, reimbursement, variability in studies regarding type and length of program, lack of professional society endorsement, and cost efficacy.⁷³ Given programs can be 2 to 3 weeks in duration and consist of simple walking programs and at-home resistance training exercises, there is little reason not to initiate such programs ubiquitously. Creative means of motivation for such programs, such as interactive phone applications, may help stimulate participation and quell skepticism.

Statement 21: Patients undergoing surgical resection for LCS-detected lesions who are actively smoking should be provided the resources and instruction to attempt to quit smoking for at least 3 weeks before surgical resection, within the 12-week treatment window.

Effective preoperative management in patients diagnosed with NSCLC detected by LCS is a critical component of optimizing surgical outcomes and long-term survival. Patients who are actively smoking at the time of diagnosis require particular consideration because tobacco use is associated with increased surgical complications, impaired healing, and poorer oncologic outcomes.^{74–76}

All patients scheduled for surgical resection of LCS-detected tumors who are actively smoking should be provided with comprehensive smoking cessation resources and clear instruction to make a dedicated attempt to quit smoking.⁷⁶ Although data on the optimal timing for smoking cessation are variable, this expert consensus panel agreed that patients be encouraged and supported to quit smoking for a minimum of 3 weeks before their procedure, while making efforts to avoid a delay in treatment beyond the recommended 12-week window. Health care teams should ensure access to a structured smoking cessation program. This should include counseling, behavioral support, and evidence-based pharmacologic aids such as nicotine replacement therapy or other approved medications.

Statement 22: In patients with LCS-detected suspected clinical stage I NSCLC with a pulmonary lesion classified as Lung Imaging Reporting

and Data System (RADS) 4A, longitudinal evaluation of lesion growth or change may be appropriate.

Lung-RADS 4A denotes a lesion with moderate suspicion for malignancy, typically carrying a 5% to 15% chance of cancer.^{77,78} Immediate intervention is not always necessary; instead, a period of longitudinal evaluation to assess lesion growth or change may be appropriate.

Short-term serial imaging with low-dose CT at 3-month intervals permits evaluation of growth dynamics and morphologic evolution. Lesions that remain stable on serial evaluation may be benign or indolent, supporting continued surveillance and sparing patients the risks of invasive procedures.⁷⁹ Conversely, interval growth or the development of high-risk features—such as spiculation, solid component increase, or metabolic activity on PET/CT—justifies proceeding with tissue diagnosis.⁸⁰

This strategy balances early detection with minimization of harm from overtreatment. It is particularly relevant in patients with significant comorbidities or limited pulmonary reserve, where procedural risks are heightened. Multidisciplinary discussion is essential to tailor management to individual risk profiles and patient preferences to determine when surveillance vs diagnostic intervention is appropriate.⁸¹ Careful patient counseling regarding the risks of cancer progression due to delayed treatment vs the risk of overtreatment is challenging yet essential.

Statement 23: The multidisciplinary LCS clinical decision-making team should, at a minimum, include thoracic surgery, medical oncology, radiation oncology, chest radiology, and pulmonology.

The role of a multidisciplinary team (tumor board) in the management of patients with lung cancer has been considered an essential component of quality care despite a historical paucity of data specific to lung cancer. The benefits of multidisciplinary team in other malignancies have shown consistent improvements in survival and increased provision of evidence-based care. Controlling for all potential variables that may impact clinical outcomes can be challenging when attempting to systematically evaluate the impact of multidisciplinary boards on overall and cancer-specific survival, and randomized trials specific to this issue are not available. However, more contemporary data on lung cancer multidisciplinary boards have provided sufficient evidence to support the recommendation for the routine use of these committees in the evaluation

of patients with biopsy specimen-proven lung cancer or suspicious lung nodules. The board should include representation from thoracic surgery, medical oncology, radiation oncology, radiology, and pulmonology.

Early studies in stage III and IV disease demonstrated mixed overall survival rates between patients presented at a multidisciplinary tumor board vs those that were not; however, even these earlier studies outlined a benefit for presented patients in receipt of the appropriate systemic therapy regimen and better concordance with recommended clinical pathways.^{81–83} More contemporary data across all stages of NSCLC have demonstrated consistent improvements in overall survival with inclusion in an multidisciplinary team (tumor board) among propensity-matched groups in both retrospective and prospective registry studies.^{84–87}

Gaudioso and colleagues⁸⁸ examined a broader group of patients more inclusive of early-stage disease and demonstrated improved overall survival (36.9 months vs 19.3 months; $P < .001$) and cancer-specific survival (48 months vs 28.1 months; $P < .001$) across all stages; however, subgroup analysis suggested similar survival between cohorts for patients with stage I/II disease, with maintenance of improved survival among patients with stage III/IV disease. Within this study, presentation at a multidisciplinary meeting (MDM) resulted in significant treatment plan changes in 41% of patients. Furthermore, compliance rates of treatment plans with National Comprehensive Cancer Network guidelines improved from 80% to 94% ($P < .001$) with MDM presentation.

A larger cohort study demonstrated improved survival across all stages of lung cancer, including stage I (hazard ratio, 1.43; 95% CI 0.88–2.32; $P < .001$) and II (hazard ratio, 2.12; 95% CI, 1.43–3.15; $P < .001$) disease.⁸⁴ In a retrospective cohort study limited to surgically treated patients, 1-year mortality was significantly lower in the MDM group vs controls (8% vs 18%, $P = .006$) with improvements in complete preoperative staging evaluation.⁸⁷

In a cohort of 9600 patients with 5900 presented at MDM, non-MDM patients were less likely to be offered treatment, including a lower proportion receiving surgery, radiotherapy, and chemotherapy.⁸⁶ Furthermore, risk factors for failing to present patients at MDM included advanced age, poor performance status, advanced clinical stage, and presentation at a smaller medical center. This suggests the

potential for bias resulting in failure to offer available treatment to some populations that may have limited access to care.

COMMENT

The implementation of low-dose CT LCS has transformed the landscape of lung cancer medicine and thoracic surgical practice by increasing the detection of lung nodules, many of which require diagnostic or therapeutic intervention. With the growing volume of screening, multidisciplinary teams are increasingly faced with decisions regarding biopsy, surgical resection, and perioperative management. Although the mortality benefit of screening is well established, concerns regarding downstream harms—including unnecessary invasive procedures, procedural morbidity, and overtreatment—have persisted. These concerns are in part fueled by misinterpretation of outcomes data as well as by variability in practice patterns and underscoring the urgent need for consensus-based standards to guide biopsy and surgical management of screen-detected nodules.

The consensus statements presented in this document provide practical benchmarks intended to promote consistency, safeguard quality, and reinforce provider and patient confidence in LCS programs. Key areas of focus include defining when tissue diagnosis is required before surgery, clarifying circumstances in which surgical resection without prior biopsy may be appropriate, and establishing acceptable complication thresholds for both diagnostic and therapeutic interventions. Equally important, these statements highlight the importance of timely intervention, balanced with the need for preoperative smoking cessation, “prehab,” and the critical role of multidisciplinary collaboration in ensuring optimal patient outcomes. Together, these benchmarks are designed to serve as a framework for clinicians, institutions, and health systems to evaluate the performance of their screening program and identify opportunities for continued quality improvement.

Of note, this ECD intentionally limited the therapeutic interventions considered to surgical lung resection. This multidisciplinary expert panel acknowledges that SBRT is an appropriate therapeutic option for selected patients with screen-detected lung nodules who are poor operative candidates or who prefer nonsurgical treatment. No completed prospective randomized trials that directly compare SBRT with surgical resection in operable patients have been published.⁸⁹ In the absence of completed,

prospective randomized trials, pulmonary resection remains the standard therapy for operable, screen-detected NSCLCs. Accordingly, our consensus statements are focused on surgical therapy.

This document also recognizes the dynamic landscape of LCS. As investigators continue to study broadening LCS eligibility criteria, such as the potential adoption of total years of smoking rather than pack-year thresholds, elimination of quit-time restrictions, and consideration of higher-risk never-smokers, the absolute number of patients undergoing biopsy or surgical resection after LCS will rise.^{6,7} Without standardized approaches, variability in management may be amplified, potentially undermining the survival benefit of screening and perpetuating perceptions of harm. Establishing clear, consensus-based standards is therefore essential to ensuring that expanded screening efforts translate into durable improvements in lung cancer outcomes.

A critical determinant of success in LCS lies not only in establishing benchmarks and ensuring quality for surgical or diagnostic interventions but also in the rigor of the screening process itself. Adherence to structured reporting systems, such as Lung-RADS, use of multidisciplinary evaluation, and ensuring compliance with longitudinal follow-up are essential safeguards against premature or unnecessary invasive procedures. Data from large contemporary programs demonstrate that when these principles are applied, the incidence of surgery or invasive procedures for benign disease is exceedingly low—less than 0.5%.^{3,35}

The overarching goal must be to maximize the cancer detection rate while simultaneously minimizing interventions for benign disease. Achieving this balance requires multidisciplinary oversight, radiologic-pathologic correlation, and an emphasis on evidence-based follow-up pathways. Only by embedding these elements into routine clinical practice can screening programs preserve the mortality benefit of LCS while addressing lingering concerns about procedural harm and overdiagnosis.

Several limitations must be acknowledged. The available evidence on biopsy and surgical management of screen-detected nodules is limited and heterogeneous, with relatively few high-quality comparative studies. Accordingly, these statements are based on the best available data from a comprehensive literature review combined with expert consensus, developed through a structured Delphi process. As new evidence

emerges, particularly from prospective studies and registries, these recommendations should be revisited and refined. Furthermore, although the consensus group represented diverse expertise, clinical disciplines, practice environments, and institutional perspectives, regional and international variation in resources and patient populations may affect implementation.

Despite these limitations, we believe that the document represents an important step toward standardizing the surgical and diagnostic management of screen-detected lung nodules. By providing clear, evidence-informed benchmarks, these statements are intended to support thoracic surgeons, multidisciplinary teams, and LCS programs in delivering safe, effective, and patient-centered care.

Looking forward, further research is needed to validate these consensus benchmarks across diverse practice settings and to assess their impact on patient outcomes. Large-scale prospective studies and registry-based analyses should aim to refine procedural thresholds, complication benchmarks, and timelines metrics. Additionally, integration of emerging technologies, such as molecular diagnostics, artificial intelligence-driven image analysis, and risk stratification, and novel minimally invasive biopsy and operative approaches may provide opportunities to further optimize management of screen-detected nodules. Continued research and innovation will be essential to ensure that these benchmarks evolve with contemporary evidence and remain responsive to the needs of patients and providers.

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REFERENCES

- Aberle DR, Adams AM, Berg CD, et al. Reduced lung-cancer mortality with low-dose computed tomographic screening. *N Engl J Med*. 2011;365:395–409. <https://doi.org/10.1056/NEJMoa102873>
- de Koning HJ, van der Aalst CM, de Jong PA, et al. Reduced lung-cancer mortality with volume CT screening in a randomized trial. *N Engl J Med*. Feb 6 2020;382:503–513. <https://doi.org/10.1056/NEJMoa1911793>
- McKee BJ, Regis S, Borondy-Kitts AK, et al. NCCN Guidelines as a model of extended criteria for lung cancer screening. *J Natl Compr Canc Netw*. 2018;16:444–449. <https://doi.org/10.6004/jnccn.2018.7021>
- Rendle KA, Saia CA, Vachani A, et al. Rates of downstream procedures and complications associated with lung cancer screening in routine clinical practice: a retrospective cohort study. *Ann Intern Med*. 2024;177:18–28. <https://doi.org/10.7326/m23-0653>
- Kamel MK, Kariyawasam S, Stiles B. Overestimation of screening-related complications in the National Lung Screening Trial. *J Thorac Cardiovasc Surg*. 2023;166:336–344.e2. <https://doi.org/10.1016/j.jtcvs.2022.10.051>
- Potter AL, Guo Q, Srinivasan D, et al. Assessing lung cancer screening eligibility of patients with lung cancer in the Boston Lung Cancer Study: an analysis of 7186 lung cancer cases. *Ann Thorac Surg*. 2025;119:768–776. <https://doi.org/10.1016/j.athoracsur.2025.01.003>
- Chang GC, Chiu CH, Yu CJ, et al. Low-dose CT screening among never-smokers with or without a family history of lung cancer in Taiwan: a prospective cohort study. *Lancet Respir Med*. 2024;12:141–152. [https://doi.org/10.1016/s2213-2600\(23\)00338-7](https://doi.org/10.1016/s2213-2600(23)00338-7)
- Feldner K, Dutka P. Exploring the evidence: generating a research question: using the PICOT framework for clinical inquiry. *Nephrol Nurs J*. 2024;51:393–395.
- Shamseer L, Moher D, Clarke M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. *BMJ*. 2015;350:g7647. <https://doi.org/10.1136/bmj.g7647>
- Kuehn BM. IOM sets out “gold standard” practices for creating guidelines, systematic reviews. *JAMA*. 2011;305:1846–1848. <https://doi.org/10.1001/jama.2011.597>
- Bakaeen FG, Svensson LG, Mitchell JD, Keshavjee S, Patterson GA, Weisel RD. The American Association for Thoracic Surgery/Society of Thoracic Surgeons position statement on developing clinical practice documents. *J Thorac Cardiovasc Surg*. 2017;153:999–1005.
- Mazzone PJ, Lam L. Evaluating the patient with a pulmonary nodule: a review. *JAMA*. 2022;327:264–273. <https://doi.org/10.1001/jama.2021.24287>
- Gould MK, Donington J, Lynch WR, et al. Evaluation of individuals with pulmonary nodules: when is it lung cancer? Diagnosis and management of lung cancer, 3rd ed: American College of Chest Physicians evidence-based clinical practice guidelines. *Chest*. 2013;143(suppl):e935–e1205. <https://doi.org/10.1378/chest.12-2351>
- Callister ME, Baldwin DR, Akram AR, et al. British Thoracic Society guidelines for the investigation and management of pulmonary nodules. *Thorax*. 2015;70(suppl 2):ii1–ii54. <https://doi.org/10.1136/thoraxjnl-2015-207168>
- Kunin JR, Flors L, Hamid A, Fuss C, Sauer D, Walker CM. Thoracic endemic fungi in the United States: importance of patient location. *Radiographics*. 2021;41:380–398. <https://doi.org/10.1148/rj.2021200071>
- Paul S, Isaacs AJ, Treasure T, Altorki NK, Sedrakyan A. Long term survival with thoracoscopic versus open lobectomy: propensity matched comparative analysis using SEER–Medicare database. *BMJ*. 2014;349:g5575. <https://doi.org/10.1136/bmj.g5575>
- Yang CF, Sun Z, Speicher PJ, et al. Use and outcomes of minimally invasive lobectomy for stage I non-small cell lung cancer in the National Cancer Data Base. *Ann Thorac Surg*. 2016;101:1037–1042. <https://doi.org/10.1016/j.athoracsur.2015.11.018>
- Lim E, Batchelor TJP, Dunning J, et al. Video-assisted thoracoscopic or open lobectomy in early-stage lung cancer. *NEJM Evid*. 2022;1:EVI-Doa2100016. <https://doi.org/10.1056/EVIDo2100016>
- Potter AL, Spasojevic A, Raman V, et al. The increasing adoption of minimally invasive lobectomy in the United States. *Ann Thorac Surg*. 2023;116:222–229. <https://doi.org/10.1016/j.athoracsur.2022.09.032>
- Rodriguez-Quintero JH, Elbahrawy MM, Montal AM, et al. Minimally invasive surgery for clinical T4 non-small-cell lung cancer: national trends and outcomes [published correction in *Eur J Cardiothorac Surg*. 2024;65:ezae203]. *Eur J Cardiothorac Surg*. 2024;65:ezae009. <https://doi.org/10.1093/ejcts/ezae009>
- Servais EL, Blasberg JD, Brown LM, et al. The Society of Thoracic Surgeons General Thoracic Surgery Database: 2022 update on outcomes and research. *Ann Thorac Surg*. 2023;115:43–49. <https://doi.org/10.1016/j.athoracsur.2022.10.025>
- Yun J, Choi YS, Hong TH, et al. Nononcologic mortality after pneumonectomy compared to lobectomy. *Semin Thorac Cardiovasc Surg*. 2022;34:1122–1131. <https://doi.org/10.1053/j.semtcvs.2021.07.014>
- Jones GD, Caso R, Tan KS, et al. Propensity-matched analysis demonstrates long-term risk of respiratory and cardiac mortality after pneumonectomy compared with lobectomy for lung cancer. *Ann Surg*. 2022;275:793–799. <https://doi.org/10.1097/sla.0000000000004065>
- El Alam R, Byrne SC, Hammer MM. Rate of benign nodule resection in a lung cancer screening program. *Clin Imaging*. 2023;104:109984. <https://doi.org/10.1016/j.clinimag.2023.109984>
- Flores R, Bauer T, Aye R, et al. Balancing curability and unnecessary surgery in the context of computed tomography screening for lung cancer. *J Thorac Cardiovasc Surg*. 2014;147:1619–1626. <https://doi.org/10.1016/j.jtcvs.2013.11.001>
- Balata H, Ruparel M, O'Dowd E, et al. Analysis of the baseline performance of five UK lung cancer screening programmes. *Lung Cancer*. 2021;161:136–140. <https://doi.org/10.1016/j.lungcan.2021.09.012>

27. Manyak A, Seaburg L, Bohrer K, Kirtland SH, Hubka M, Gerbino AJ. Invasive procedures associated with lung cancer screening in clinical practice. *Chest*. 2023;164:544–555. <https://doi.org/10.1016/j.chest.2023.02.010>
28. Gagné S, McIsaac DI. Modifiable risk factors for patients undergoing lung cancer surgery and their optimization: a review. *J Thorac Dis*. 2018;10(suppl 32):S3761–S3772. <https://doi.org/10.21037/jtd.2018.10.04>
29. Sancheti MS, Melvan JN, Medbery RL, et al. Outcomes after surgery in high-risk patients with early stage lung cancer. *Ann Thorac Surg*. 2016;101:1043–1051.
30. Lentz RJ, Frederick-Dyer K, Planz VB, et al. Navigational bronchoscopy or transthoracic needle biopsy for lung nodules. *N Engl J Med*. 2025;392:2100–2112.
31. Roy V, Ugalde PA, Bourdages-Pageau E, Lacasse Y, Labbé C. Trans-thoracic needle biopsy versus surgical diagnosis for solid pulmonary nodules. *J Thorac Dis*. 2022;14:2472–2480.
32. Mohamed A, Donthi D, Malik P, et al. Diagnostic value of intra-operative fine-needle aspiration cytologic diagnosis versus frozen section-based histopathologic diagnosis in thoracic malignancies: a single institution's experience. *Diagn Cytopathol*. 2023;51:589–595.
33. van't Westeinde SC, Horeweg N, De Leyn P, et al. Complications following lung surgery in the Dutch–Belgian randomized lung cancer screening trial. *Eur J Cardiothorac Surg*. 2012;42:420–429. <https://doi.org/10.1093/ejcts/ezs081>
34. Archer JM, Mendoza DP, Hung YP, Lanuti M, Digumarthy SR. Surgical resection of benign nodules in lung cancer screening: incidence and features. *JTO Clin Res Rep*. 2023;4:100605. <https://doi.org/10.1016/j.jtccr.2023.100605>
35. Ho H, Williamson C, Regis SM, et al. Surgery and invasive diagnostic procedures for benign disease are rare in a large low-dose computed tomography lung cancer screening program. *J Thorac Cardiovasc Surg*. 2021;161:790–802.e2. <https://doi.org/10.1016/j.jtcvs.2020.08.109>
36. Crosbie PA, Balata H, Evison M, et al. Second round results from the Manchester 'Lung Health Check' community-based targeted lung cancer screening pilot. *Thorax*. 2019;74:700–704. <https://doi.org/10.1136/thoraxjnl-2018-212547>
37. Fernandez-Bussy S, Chandra NC, Koratala A, et al. Robotic-assisted bronchoscopy: a narrative review of systems. *J Thorac Dis*. 2024;16:5422–5434.
38. Bhandari BS, Jain A, Sharma S, Rana G, Sabath BF. Robotic bronchoscopy: a comprehensive review. *J Respir*. 2024;4:128–139.
39. Pathak K, Hart S, Sutter J, Meyer T, Ross P, Skabla P. Robotic navigation bronchoscopy using the Ion platform: a new era in proving benign disease. *CHEST*. 2021;160:A2052. <https://doi.org/10.1016/j.chest.2021.07.1812>
40. Tammemägi MC, Borondy-Kitts A, Field JK, et al. Lung cancer screening program quality indicators—review and recommendations: an international association for the Study of Lung Cancer Delphi Process Study. *J Thorac Oncol*. 2025;20:871–883. <https://doi.org/10.1016/j.jtho.2025.01.019>
41. Lang D, Reinelt V, Horner A, et al. Complications of CT-guided transthoracic lung biopsy: a short report on current literature and a case of systemic air embolism. *Wien Klin Wochenschr*. 2018;130:288–292. <https://doi.org/10.1007/s00508-018-1317-0>
42. Deng CJ, Dai FQ, Qian K, et al. Clinical updates of approaches for biopsy of pulmonary lesions based on systematic review. *BMC Pulm Med*. 2018;18:146. <https://doi.org/10.1186/s12890-018-0713-6>
43. Folch EE, Bowling MR, Pritchett MA, et al. NAVIGATE 24-month results: electromagnetic navigation bronchoscopy for pulmonary lesions at 37 centers in Europe and the United States. *J Thorac Oncol*. 2022;17:519–531. <https://doi.org/10.1016/j.jtho.2021.12.008>
44. Núñez ER, Caverly TJ, Zhang S, et al. Invasive procedures and associated complications after initial lung cancer screening in a national cohort of veterans. *Chest*. 2022;162:475–484. <https://doi.org/10.1016/j.chest.2022.02.031>
45. Pyrali FF, Hakami-Majd N, Sabbahi W, Chaux G. Robotic-assisted navigation bronchoscopy: a meta-analysis of diagnostic yield and complications. *J Bronchology Interv Pulmonol*. 2024;31:70–81. <https://doi.org/10.1097/LBR.0000000000000942>
46. Verhoeven RLJ, Fütterer JJ, Hoefsloot W, van der Heijden E. Cone-beam CT image guidance with and without electromagnetic navigation bronchoscopy for biopsy of peripheral pulmonary lesions. *J Bronchology Interv Pulmonol*. 2021;28:60–69. <https://doi.org/10.1097/LBR.0000000000000697>
47. Styrvoky K, Schwalk A, Pham D, Madsen K, Chiu HT, Abu-Hijleh M. Radiation dose of cone beam CT combined with shape sensing robotic assisted bronchoscopy for the evaluation of pulmonary lesions: an observational single center study. *J Thorac Dis*. 2023;15:4836–4848. <https://doi.org/10.21037/jtd-23-587>
48. Shaller BD, Duong DK, Swenson KE, Free D, Bedi H. Added value of a robotic-assisted bronchoscopy platform in cone beam computed tomography-guided bronchoscopy for the diagnosis of pulmonary parenchymal lesions. *J Bronchology Interv Pulmonol*. 2024;31:2024;31:e0971. <https://doi.org/10.1097/LBR.0000000000000971>
49. Katsis J, Roller L, Aboudara M, et al. Diagnostic yield of digital tomosynthesis-assisted navigational bronchoscopy for indeterminate lung nodules. *J Bronchology Interv Pulmonol*. 2021;28:255–261. <https://doi.org/10.1097/LBR.0000000000000766>
50. Low SW, Lentz RJ, Chen H, et al. Shape-sensing robotic-assisted bronchoscopy vs digital tomosynthesis-corrected electromagnetic navigation bronchoscopy: a comparative cohort study of diagnostic performance. *Chest*. 2023;163:977–984. <https://doi.org/10.1016/j.chest.2022.10.019>
51. Kozower BD, Sheng S, O'Brien SM, et al. STS database risk models: predictors of mortality and major morbidity for lung cancer resection. *Ann Thorac Surg*. 2010;90:875–883.
52. Hou B, Deng XF, Zhou D, Liu QX, Dai JG. Segmentectomy versus wedge resection for the treatment of high-risk operable patients with stage I non-small cell lung cancer: a meta-analysis. *Ther Adv Respir Dis*. 2016;10:435–443. <https://doi.org/10.1177/1753465816667121>
53. Altorki N, Wang X, Kozono D, et al. Lobar or sublobar resection for peripheral stage IA non-small-cell lung cancer. *N Engl J Med*. 2023;388:489–498. <https://doi.org/10.1056/NEJMoa2212083>
54. Merritt RES JB, Wernly JA. Outcomes of segmentectomy versus wedge resection for clinical stage I NSCLC using ACS NSQIP data. *J Thorac Cardiovasc Surg*. 2022;164:421–432.
55. Kneuert PJ, Zhao J, D'Souza DM, Abdel-Rasoul M, Merritt RE. National trends and outcomes of segmentectomy in the Society of Thoracic Surgeons Database. *Ann Thorac Surg*. 2022;113:1361–1369.
56. Towle CW, Servais EL, Brown LM, et al. The Society of Thoracic Surgeons General Thoracic Surgery Database: 2023 update on outcomes and research. *Ann Thorac Surg*. 2024;117:489–496.
57. Altorki NK, Wang X, Wigle D, et al. Perioperative mortality and morbidity after sublobar versus lobar resection for early-stage non-small-cell lung cancer: post-hoc analysis of an international, randomized, phase 3 trial (CALGB/Alliance 140503). *Lancet Respir Med*. 2018;6:915–924. [https://doi.org/10.1016/S2213-2600\(18\)30411-9](https://doi.org/10.1016/S2213-2600(18)30411-9)
58. Krantz SB, Mitzman B, Antonoff MB, et al. Thoracic Surgery Outcomes Research Network (ThORN) consensus document on defining a high-quality wedge resection for early-stage lung cancer. *Ann Thorac Surg*. 2025;119:944–956. <https://doi.org/10.1016/j.athoracsur.2024.12.017>
59. Fernandez FG, Kosinski AS, Burfeind W, et al. The Society of Thoracic Surgeons lung cancer resection risk model: higher quality data and superior outcomes. *Ann Thorac Surg*. 2016;102:370–377.
60. Boffa DJ, Dhamija A, Kosinski AS, et al. Fewer complications result from a video-assisted approach to anatomic resection of clinical stage I lung cancer. *J Thorac Cardiovasc Surg*. 2014;148:637–643.
61. Yang C-F, Kumar A, Deng JZ, et al. A national analysis of short-term outcomes and long-term survival following thorascopic versus open lobectomy for clinical stage II non-small-cell lung cancer. *Ann Surg*. 2021;273:595–605.

62. Yang H-X, Woo KM, Sima CS, et al. Long-term survival based on the surgical approach to lobectomy for clinical stage I nonsmall cell lung cancer: comparison of robotic, video-assisted thoracic surgery, and thoracotomy lobectomy. *Ann Surg*. 2017;265:431–437.
63. Burt BM, Kosinski AS, Shrager JB, Onaitis MW, Weigel T. Thoracoscopic lobectomy is associated with acceptable morbidity and mortality in patients with predicted postoperative forced expiratory volume in 1 second or diffusing capacity for carbon monoxide less than 40% of normal. *J Thorac Cardiovasc Surg*. 2014;148:19–29.e1.
64. Hall H, Tocock A, Burdett S, et al. Association between time-to-treatment and outcomes in non-small cell lung cancer: a systematic review. *Thorax*. 2022;77:762–768. <https://doi.org/10.1136/thoraxjnl-2021-216865>
65. Tang A, Ahmad U, Raja S, et al. How much delay matters? How time to treatment impacts overall survival in early stage lung cancer. *Ann Surg*. 2023;277:e941–e947.
66. Banks KC, Dusendang JR, Schmittiel JA, et al. Association of surgical timing with outcomes in early stage lung cancer. *World J Surg*. 2023;47:1323–1332.
67. Heiden BT, Eaton DB Jr, Engelhardt KE, et al. Analysis of delayed surgical treatment and oncologic outcomes in clinical stage I non-small cell lung cancer. *JAMA Network Open*. 2021;4:e2111613. <https://doi.org/10.1001/jamanetworkopen.2021.11613>
68. Heiden BT, Eaton DB Jr, Chang SH, et al. Association between surgical quality metric adherence and overall survival among US veterans with early-stage non-small cell lung cancer. *JAMA Surg*. 2023;158:293–301. <https://doi.org/10.1001/jamasurg.2022.6826>
69. Tupper HI, Sarovar V, Banks KC, et al. Time to surgery in early-stage non-small cell lung cancer: defining the optimal diagnosis-to-resection interval to reduce mortality. *J Thorac Cardiovasc Surg*. 2025;169:1563–1572.e5. <https://doi.org/10.1016/j.jtcvs.2024.11.040>
70. Granger C, Cavalheri V. Preoperative exercise training for people with non-small cell lung cancer. *Cochrane Database Syst Rev*. 2022;9:CD012020. <https://doi.org/10.1002/14651858.CD012020.pub3>
71. Cavalheri V, Granger C. Preoperative exercise training for patients with non-small cell lung cancer. *Cochrane Database Syst Rev*. 2017;6:CD012020. <https://doi.org/10.1002/14651858.CD012020.pub2>
72. Brat K, Sova M, Homolka P, et al. Multimodal prehabilitation before lung resection surgery: a multicentre randomised controlled trial. *Br J Anaesth*. 2025;135:188–196. <https://doi.org/10.1016/j.bja.2025.03.018>
73. Sanchez-Lorente D, Navarro-Ripoll R, Guzman R, et al. Prehabilitation in thoracic surgery. *J Thorac Dis*. 2018;10(suppl 22):S2593–S2600. <https://doi.org/10.21037/jtd.2018.08.18>
74. Sørensen LT. Wound healing and infection in surgery: the pathophysiological impact of smoking, smoking cessation, and nicotine replacement therapy: a systematic review. *Ann Surg*. 2012;255:1069–1079.
75. Nakagawa M, Tanaka H, Tsukuma H, Kishi Y. Relationship between the duration of the preoperative smoke-free period and the incidence of postoperative pulmonary complications after pulmonary surgery. *Chest*. 2001;120:705–710.
76. Mason DP, Subramanian S, Nowicki ER, et al. Impact of smoking cessation before resection of lung cancer: a Society of Thoracic Surgeons General Thoracic Surgery Database study. *Ann Thorac Surg*. 2009;88:362–371.
77. Silvestri GA, Goldman L, Tanner NT, et al. Outcomes from more than 1 million people screened for lung cancer with low-dose CT imaging. *Chest*. 2023;164:241–251.
78. Kocher Wulfeck M, Plesner S, Herndon JE 2nd, Christensen JD, Patz EF Jr. Characterizing Lung-RADS category 4 lesions in a university lung cancer screening program. *Lung Cancer*. 2023;186:107420.
79. Jonas DE, Reuland DS, Reddy SM, et al. Screening for lung cancer with low-dose computed tomography: updated evidence report and systematic review for the US Preventive Services Task Force. *JAMA*. 2021;325:971–987. <https://doi.org/10.1001/jama.2021.0377>
80. Wood DE, Kazerooni EA, Aberle DR, et al. NCCN Guidelines® Insights: Lung Cancer Screening, Version 1.2025. *J Natl Compr Canc Netw*. 2025;23:e250002. <https://doi.org/10.6004/jnccn.2025.0002>
81. Boxer MM, Vinod SK, Shafiq J, Duggan KJ. Do multidisciplinary team meetings make a difference in the management of lung cancer? *Cancer*. 2011;117:5112–5520. <https://doi.org/10.1002/cncr.26149>
82. Friedman EL, Krukltis RJ, Patson BJ, Sopka DM, Weiss MJ. Effectiveness of a thoracic multidisciplinary clinic in the treatment of stage III non-small-cell lung cancer. *J Multidiscip Healthc*. 2016;9:267–274. <https://doi.org/10.2147/jmdh.S98345>
83. Pan CC, Kung PT, Wang YH, Chang YC, Wang ST, Tsai WC. Effects of multidisciplinary team care on the survival of patients with different stages of non-small cell lung cancer: a national cohort study. *PLoS One*. 2015;10:e0126547. <https://doi.org/10.1371/journal.pone.0126547>
84. Stone E, Rankin N, Kerr S, et al. Does presentation at multidisciplinary team meetings improve lung cancer survival? Findings from a consecutive cohort study. *Lung Cancer*. 2018;124:199–204. <https://doi.org/10.1016/j.lungcan.2018.07.032>
85. Bilfinger TV, Albano D, Perwaiz M, Keresztes R, Nemesure B. Survival outcomes among lung cancer patients treated using a multidisciplinary team approach. *Clin Lung Cancer*. 2018;19:346–351. <https://doi.org/10.1016/j.clcc.2018.01.006>
86. Lin T, Pham J, Paul E, et al. Impacts of lung cancer multidisciplinary meeting presentation: drivers and outcomes from a population registry retrospective cohort study. *Lung Cancer*. 2022;163:69–76. <https://doi.org/10.1016/j.lungcan.2021.12.006>
87. Tamburini N, Maniscalco P, Mazzara S, et al. Multidisciplinary management improves survival at 1 year after surgical treatment for non-small-cell lung cancer: a propensity score-matched study. *Eur J Cardiothorac Surg*. 2018;53:1199–1204. <https://doi.org/10.1093/ejcts/ezx464>
88. Gaudioso C, Sykes A, Whalen PE, et al. Impact of a thoracic multidisciplinary conference on lung cancer outcomes. *Ann Thorac Surg*. 2022;113:392–398. <https://doi.org/10.1016/j.athoracsur.2021.03.017>
89. Udelsman BV, Canavan ME, Zhan PL, et al. Overall survival in low-comorbidity patients with stage I non-small cell lung cancer who chose stereotactic body radiotherapy compared to surgery. *J Thorac Cardiovasc Surg*. 2024;167:822–833.e7. <https://doi.org/10.1016/j.jtcvs.2023.07.021>

Optimizing Lung Cancer Screening



INVITED COMMENTARY:

Fifteen years after the National Lung Screening Trial showed that low-dose computed tomography

screening of individuals with a current or former history of smoking was associated with a reduction in lung cancer mortality, this modality remains highly underutilized. The United States lacks a coordinated national program that could